SOME/IP Intrusion Detection System Using Real-Time and Retroactive Anomaly Detection

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***Abstract*—As the amount of in-vehicle data and number of in-vehicle functions have increased, automotive Ethernet and IP-based protocols have been adopted in vehicles. The scalable service-oriented middleware over IP (SOME/IP) protocol is designed for sending periodic or direct control messages and for updating or appending scalable functions. International regulation UN R-155 requires security monitoring to analyze and detect cyber-threats, vulnerabilities, and cyber-attacks from vehicle data and logs. SOME/IP also should be monitored from the perspective of security. We report a classification of use cases and possible attack patterns for SOME/IP, and we propose an intrusion detection system (IDS) that achieves high accuracy by combining two whitelist-based anomaly detection algorithms. The first is a real-time algorithm that compares received SOME/IP packets with a normal communication model to determine whether they are anomalies. The attack patterns covered in this paper include sophisticated malicious packets that are difficult for the real-time detection algorithm to distinguish from normal packets. Hence, the second algorithm performs retroactive detection by determining whether packets are anomalous from long-term time characteristics spanning an entire SOME/IP session. Although the retroactive algorithm takes longer for detection, we confirm a high level of detection accuracy for the proposed IDS, with a true positive rate of 0.91 and a false positive rate of 0.052.**

Keywords—vehicle security, automotive Ethernet, SOME/IP, intrusion detection system, anomaly detection

I. Introduction

The standardization and adoption of Ethernet as an in-vehicle communication protocol between electronic control units (ECUs) have advanced with the development of in-vehicle functions and the increase in the amount of communication data [1]. Automotive Ethernet communication is expected to be used in a wide range of applications, such as communication of control and sensor information (e.g., for cameras), or for conventional control communication protocols among ECUs such as controller area network (CAN) [2]. Miller et al. [3] showed that a vehicle can be controlled via unauthorized CAN communications transmitted by an attacker who illegally enters the in-vehicle network remotely from an out-of-vehicle network. This poses the same threat to in-vehicle Ethernet. International regulation UN R-155 requires security monitoring to analyze and detect cyber-threats, vulnerabilities, and cyber-attacks from vehicle data and logs. Thus, anomaly detection technology to detect attacks is an important countermeasure for in-vehicle Ethernet. In this paper, we propose an intrusion detection system (IDS) for the scalable service-oriented middleware over Internet protocol (SOME/IP), which is designed to transmit important vehicle control in Ethernet communications as a substitute for technologies such as the CAN.

Previous research [4] evaluated attacks in which unauthorized packets are inserted into normal SOME/IP communications or packets are delayed by some means. For a man-in-the-middle (MITM) attack or a case when only an attacker and a victim ECU communicate, common attack patterns in IP communication have not been evaluated. Zelle et al. [5] were the first to show that an MITM attack can be implemented in SOME/IP. However, they proposed a preventive measure by adding an authentication function to the communication protocol but did not propose an anomaly detection technique.

In this paper, we classify possible use cases and attack patterns for SOME/IP. For normal communication with existing control packets like in the CAN, we examine how SOME/IP can be used, and we classify the resulting patterns. We also evaluate attack patterns by strong attackers, which the existing research has not addressed. Specifically, there are cases in which an attacker can communicate with parties other than the original transmission/reception party. This can be done by falsifying network settings such as the address resolution protocol (ARP) table of each in-vehicle device, with communication between the attacker and victim ECU by simply interrupting a normal ECU's communication, and with SOME/IP packets that can likely be transmitted by knowing the target vehicle's communication specifications. Hence, we implement these attacks in a verification environment and prepare anomalies, including normal and attack packets that cause anomalous operation in the environment.

The proposed IDS detects the above attacks for the SOME/IP and SOME/IP-service discovery (SD) protocols. To the best of our knowledge, as there are no real SOME/IP attacks, we use whitelist-based anomaly detection. To achieve high accuracy, we combine two detection algorithms. First, we use a per-packet algorithm to detect anomalies in real time. This algorithm compares observed packets to a normal communication model on a packet-by-packet basis. However, it may overlook cases when anomalous packets cannot be differentiated from normal packets. Thus, we also implement a retroactive detection algorithm that finds anomalies in a series of packets. The retroactive detection algorithm improves the total detection accuracy but requires more time because it observes numerous packets. Because features in a layer lower than SOME/IP may be disguised by a strong attacker, we generate a normal communication model from features in the SOME/IP layer and adopt an algorithm for judging packet anomalies.

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We evaluated the proposed IDS's detection performance with datasets based on our classification of use cases and attack patterns. The IDS had a true positive rate (TPR) of approximately 0.91 and a false positive rate (FPR) of approximately 0.052. In summary, the main contributions of this research are as follows.

- We classify normal communication patterns and attack patterns for SOME/IP in use cases such as conventional communication in the CAN. We examine more powerful and feasible attacks than in existing IDS research and confirm that these attacks can be implemented in an experimental verification environment.
- We propose a whitelist-based anomaly detection system that combines two algorithms for real-time and retroactive detection of anomalous SOME/IP packets. We then evaluate the detection performance in the above environment.

II. Related Work

A. Security Research on SOME/IP

Iorio et al. [6] proposed a secure handshake and encrypted-communication method for SOME/IP. Lauser et al. [7] used a formal approach to evaluate the security of AUTOSAR SecOC, a signature standard that is also available for SOME/IP. Kreissl [8] showed that SOME/IP is vulnerable to spoofing, eavesdropping, replay, and denial-of-service attacks. Li et al. [9] proposed a fuzzer for SOME/IP that inserts an anomalous communication pattern in which the packet payload randomly changes. The vulnerability of ECU software has thus been investigated, but not for the SOME/IP protocol.

Regarding IDSs for SOME/IP, Herold et al. [10] applied a complex event engine approach to SOME/IP and proposed an IDS for four types of anomalies. The anomaly types were (1) malformed packets; (2) protocol violations such as communication patterns that are not SOME/IP-compliant; (3) system-specific violations such as deviations from individual SOME/IP services; and (4) timing issues such as deviations in expected packet arrival times. They especially focused on the third type but did not detail specific methods for these attacks, and they did not cover SOME/IP-SD communication. Gehrmann and Duplys [4] proposed an IDS based on whitelisting and signatures among genuine hosts and servers. However, they explained no specific attacks, and their IDS's scope did not include cases like rerouting of an in-vehicle network.

B. IDS for CAN Protocol

Many IDSs have been proposed for the CAN protocol, which is still widely used in control communication. Examples include an IDS focused on communication intervals [11], an IDS for detecting anomalous payload transitions via machine learning [12], and a spoofing detection method using source ECU identification based on voltage change characteristics [13]. Because many aspects of the CAN differ from IP-based protocols—e.g., the transmission source and destination are unknown, and the physical layer is very different—these IDSs are not directly applicable to SOME/IP communication. Attacks are classified by the attacker's strength against the CAN in [13], but not for SOME/IP. We thus cannot assume that the existing CAN security research on attackers is sufficient for SOME/IP communication.

III. Overview of SOME/IP

SOME/IP is a middleware solution to transport control data over Ethernet and an automotive/embedded communication protocol that supports remote procedure calls, event notifications, and an underlying serialization/wire format [15, 16]. Use case examples include infotainment systems and driving assistance. SOME/IP-SD is an extension for locating service instances and implementing publish/subscribe handling [17].

SOME/IP has four service interfaces: Request/Response, Fire & Forget, Notification Events, and Fields. In Request/Response, a communication partner (client) sends a request packet that is answered by another communication partner (server). Fire & Forget is a method invocation with only a request packet. In Notification Events, the server publishes a service to which a client subscribes, e.g., cyclically or when a value changes. Fields represent remotely accessible properties including getter, setter, and notification. A subscriber calls a getter to explicitly query a provider for a value, and a setter to change the value on the provider side. A notification is sent when a field value changes.

In SOME/IP use cases that are similar to CAN and FlexRay communication, Request/Response can be used for service control requests and responses, and Notification Events can be used for sensor information notifications, for example. Then, Fire & Forget and Fields can be considered as subsets of Request/Response and Notification Events. We thus investigate the Request/Response and Notification Events in this paper.

IV. Target Attacks and Datasets

In this section, we describe possible attack patterns for SOME/IP's Request/Response and Notification Events interfaces. To support our evaluation of the proposed IDS, we also describe an experimental environment and datasets of normal and anomalous communication to simulate actual use cases.

A. Attack Patterns

Because SOME/IP and SOME/IP-SD do not have authentication and encryption functions, an attacker can easily obtain any server's or client's communication information (e.g., its MAC address, IP address, and service ID), enabling the attacker to send SOME/IP packets or subscribe to a service by accessing the network. Hence, we assume that the attacker analyzes and obtains such information in advance. We also assume that the attacker can send plausible packets by analyzing the communication specification. Lastly, we assume three patterns when an attack occurs: connection of the attacker’s device, remote intrusion, and node modification.

Next, we determine the attack pattern from the relative positions between normal nodes and the attacker, along with the receiver's communication state. For the relative positions, we determine two patterns, indicating MITM or other attack types. For the receiver's communication state, we also determine two patterns, indicating only packets from the attacker, or packets from both normal nodes and the attacker. Fig. 1, 2, and 3 show the results of applying these attack patterns to Notification Events, Requests, and Responses, respectively.

B. Experimental Environment

Fig. 4 shows the virtual communication environment that we developed to simulate the assumed architecture in a vehicle. It includes five ECUs to implement the attacks described

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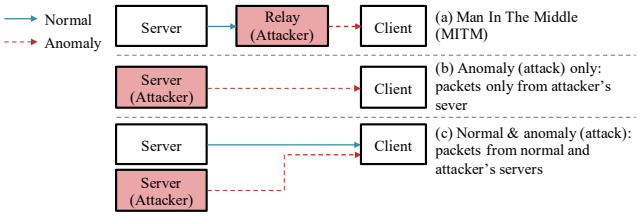

Fig. 1. Attack patterns for Notification Events.

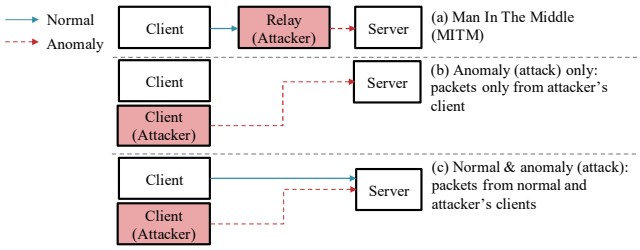

Fig. 2. Attack patterns for Requests

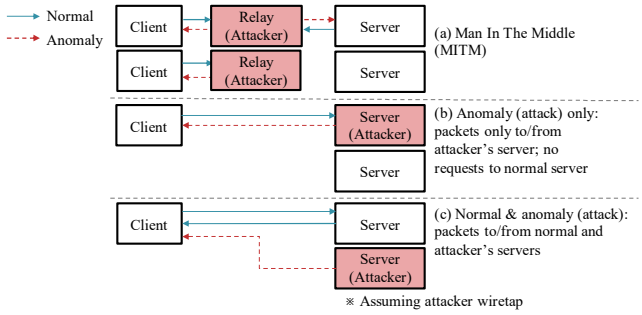

Fig. 3. Attack patterns for Responses.

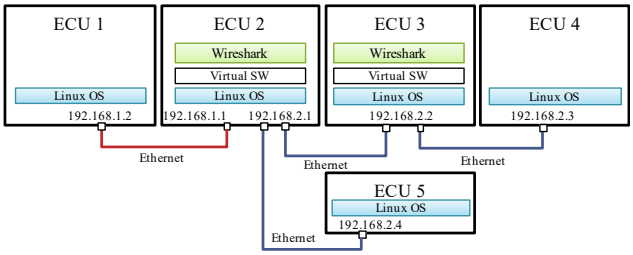

Fig. 4. Experimental network topology.

above. In this environment, multiple ECUs can simultaneously provide or subscribe to multiple services. ECUs 2 and 3 also have a routing function. VSOMEIP [18] is installed in each node, and SOME/IP and SOME/IP-SD communication uses the user datagram protocol (UDP) between nodes.

C. Datasets

We created normal and anomaly datasets for Request/Response and Notification Events. We categorized three types of Notification Events: those sent periodically by a server, those sent only when a value changes, and those sent either periodically or when a value changes. In this paper, we do not use datasets and survey detection methods for the first type of Notification Event, because works such as [19] have extensively discussed IDSs for periodic transmission. Many methods focusing on communication intervals can be used with various protocols by just changing a monitoring identifier.

As summarized in Table I, we prepared three types of normal data patterns, comprising Request/Response and two types of Notification Events, and we created anomaly datasets for them. The payloads of the normal datasets were implemented by referring to actual in-vehicle CAN communication packets. This was because certain packets such as those for driving support are conventionally communicated via the CAN but may use Ethernet instead. These packets have different meanings in bytes, and each has a range of values and a pattern of value changes. The normal datasets contain all ranges of values and patterns of changes. The anomaly datasets contain the attack patterns shown in Fig. 1, 2, and 3 for each target, as listed in Table II. The attacker communication follows the SOME/IP specification. We created a total of 25 anomaly datasets with various target payloads and attack values.

Here, we describe the correspondences between the four types of anomalies described in [10] and our anomaly datasets. We exclude malformed packets and protocol violations from the IDS detection scope because they are easily detectable, and we thus did not include them in the anomaly datasets. On the other hand, many of the anomaly datasets include system-specific violations, as well as attacks that deviate from service-specific ranges of values and patterns of value changes. Timing issues are also included in datasets in which normal and attack packets coexist. In addition, for each attack pattern, the datasets include plausible attack packets without system-specific violations or timing issues. That is, the datasets assume the possibility of strong attacks not addressed by existing research.

TABLE I.
OVERVIEW OF DATASETS

No.	Overview
Normal_RR	Normal Request/Response packets
Normal_NEpc	Normal Notification Event packets (sent periodically or when a value changes)
Normal_NEc	Normal Notification Event packets (sent when a value changes)
Anomaly_RR	Attack packets against Request/Response in Normal_RR
Anomaly_NEpc	Attack packets against Notification Events (sent periodically or when a value changes) in Normal_NEpc
Anomaly_NEc	Attack packets against Notification Events (sent when a value changes) in Normal_NEc

TABLE II.
OVERVIEW OF ANOMALY DATASETS

Target packets	(a) MITM	(b) Anomaly only	(c) Normal & anomaly
Request	Anomaly_RR ×1 set	Anomaly_RR×1 set	Anomaly_RR ×3 set
Response	Anomaly_RR ×1 set	Anomaly_RR×1 set	-
Notification Events (*)	Anomaly_NEpc ×8 sets	Anomaly_NEc ×5 sets (§)	Anomaly_NEpc ×2 sets
Notification Events (†)	Anomaly_NEc ×1 set		Anomaly_NEc ×2 sets

*: sent periodically or when a value changes.
†: sent when a value changes.
§: Anomaly_NEpc is included in the Anomaly_NEc datasets.

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In the Normal_RR dataset, two ECUs establish publish/subscribe by using SOME/IP-SD and then perform Request/Response communication by using SOME/IP. The response is either OK or NG (meaning that the server cannot accept the request). In the dataset, the service IDs used by clients are assumed to include Request/Response and Notification Events. We implement a use case in which an EventGroup to which a client subscribes does not contain an Event that is sent periodically, and no state change is required for sending. Although this dataset does not include them, there are also use cases for Request/Response communication without establishing publish/subscribe.

In the Normal_NEpc dataset, after two ECUs establish publish/subscribe, the server sends Notification Events periodically or when a value changes. The use cases include Request/Response communications that have the same service ID and different method IDs, and the state changes according to the communication contents. Depending on the state, certain bit ranges can change the value range and value-change pattern of the Notification Event payload. Additionally, the Normal_NEpc dataset includes Normal_RR communications in parallel with the communications described above.

In the Normal_NEc dataset, after two ECUs establish publish/subscribe, the server only sends Notification Events when a value changes. The other specifications are the same as for the Normal_NEpc dataset but with a different service ID. The source and destination addresses are also the same as for Normal_NEpc. The Normal_NEc dataset also includes Normal_RR and Normal_NEpc communications that exist in parallel with the communications described above.

With these normal datasets, we generated Normal_RR data containing 6,455 packets, Normal_NEpc data containing 255,128 packets, and Normal_NEc data containing 1,722,011 packets. These packets were used as the IDS learning data described in Section V.

As for the anomaly datasets, we describe the implementations of four of them here. They entail three attack patterns with Request as the target and one with Notification Events as the target. We omit descriptions of the others because they were implemented in the same way as these four.

The Anomaly_RR dataset represents an attack on Normal_RR. Here, the attacker connects a device to ECU 2, as illustrated in Fig. 5 for an MITM attack and in Fig. 6 for client impersonation. First, we describe the data for an MITM attack on Normal_RR. During SOME/IP communication between a normal client and server, the attacker rewrites their ARP tables to induce communication with the attacker through ARP spoofing. Subsequently, the attacker establishes publish/subscribe with the normal server and client through SOME/IP-SD communication. The attacker then receives a request illegally and sends a response.

Next, we describe the data for an attack on Normal_RR by using packets from both normal and attacker clients. Both the normal client and the attacker subscribe to services on a normal server via SOME/IP-SD, and both communicate via Request/Response with the normal server. The anomaly datasets for this attack include cases in which the attacker both does and does not know the pattern of value changes.

Next, we describe the data for an attack on Normal_RR in which the server only receives packets from an attacker client. Both a normal client and the attacker subscribe to a service on a normal server via SOME/IP-SD. When the normal client does not send anything, the attacker sends a request.

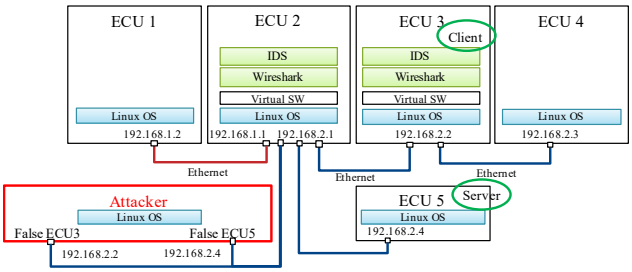

Fig. 5. Execution environment for MITM.

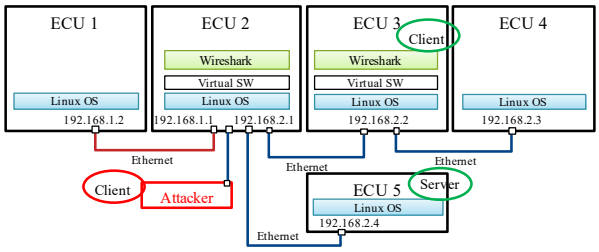

Fig. 6. Execution environment for client impersonation.

TABLE III.
OVERVIEW OF NUMBER OF PACKETS IN DATASET

Category	Attack Pattern	Anomaly/All [packet]
Anomaly_RR	(a) MITM	1,545/6,582
Anomaly_RR	(b) Anomaly only	568/5,873
Anomaly_RR	(c) Normal & Anomaly	766/3,814
Anomaly_NEpc	(a) MITM	13,858/32,623
Anomaly_NEpc	(b) Anomaly only	-
Anomaly_NEpc	(c) Normal & Anomaly	2,825/8,474
Anomaly_NEc	(a) MITM	2,313/5,385
Anomaly_NEc	(b) Anomaly only	901/4,888
Anomaly_NEc	(c) Normal & Anomaly	12,701/24,308

Finally, we describe the data for an attack on Normal_NEc in which a client only receives packets from an attacker's server. Initially, the attacker uses ARP spoofing to rewrite an invalid MAC address for a server used by the client. This prevents the client's subscribe packets from reaching the server. Then, the attacker establishes publish/subscribe with the client via SOME/IP-SD. When this attack is performed on Normal_NEc, the two types of Notification Events have the same source and destination; thus, there are no Notification Events from a legitimate server, only from the attacker.

As described above, we created the normal datasets to reflect actual use cases by involving multiple services and subscriptions simultaneously. The payloads' ranges of values and changes in value patterns covered various bit ranges, and we thus created anomaly datasets that covered the corresponding attack patterns. Table III summarizes the generated anomaly datasets and their ratios of attack packets. In the ratios, the denominator is the total number of SOME/IP and SOME/IP-SD packets, including both normal and attack packets, and the numerator is the number of attack packets.

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V. Proposed System

The proposed IDS monitors SOME/IP and SOME/IP-SD communication for each SOME/IP service. SOME/IP services are distinguished by the service ID and client ID header features. A normal communication model for each service is constructed from transaction packets acquired in advance. When a packet deviating from a normal communication model is observed during monitoring, it is determined to be anomalous. Each time a packet is received, a real-time detection algorithm, which has a micro viewpoint, inspects the packet for anomalies. SOME/IP is a publish/subscribe communication protocol; accordingly, we expect that when a packet is detected as anomalous, other packets transmitted from the same source have also to be detected as anomalous, even if the payload's value and timing are reasonable and the other packets were overlooked by the real-time detection algorithm. Hence, a retroactive detection algorithm, which has a macro viewpoint, inspects those other packets again. The retroactive algorithm also inspects packets that are overlooked by the real-time algorithm, according to the communication interval characteristics of the whole session for each service.

A. Background and Design Concept

The IDS targets attacks that are intended to control vehicles fraudulently through anomalous communication. We thus assume that the attacker's anomalous packets adhere to the SOME/IP specification and conform to appropriate communication. In other words, the proposed IDS's detection scope excludes the malformed packets and protocol violations described in [10], because they can easily be detected. Instead, the IDS targets the other two types of anomalies, i.e., system-specific violations and timing issues.

1) Architecture: The IDS is installed where it can monitor all SOME/IP packets. In other words, we assume that the in-vehicle Ethernet cables are wired so that the IDS can directly monitor all SOME/IP packets, or that there are devices such as IP switches that bypass the in-vehicle Ethernet. In the latter case, the bypass function must not be compromised.

2) Characteristics: We assume that an attacker knows the normal state of the target vehicle, including valid MAC addresses, IP addresses, service IDs, and payloads. We also assume that communications can be transmitted and received through impersonation. Accordingly, the proposed IDS uses only the observation time of each packet and the features obtained from the SOME/IP layer.

3) Implementation Cost: The retroactive detection algorithm described below inspects the normality of an entire SOME/IP service session. It thus requires temporary storage of all packet detection results for an entire session. An ECU equipped with the proposed IDS is assume to have a sufficient storage capacity.

B. Real-Time Detection Algorithm

The real-time detection algorithm entails two phases: a training phase to model the pattern of normal SOME/IP communications, and an observation and detection phase. Fig. 7 and 8 show the respective workflows of these phases. In the training phase, a received packet is compared to the immediately preceding packet in terms of the workflow, and a state transition at a certain moment is recorded. The resulting state transition pattern is used to form a normal communication model. Then, during the observation and detection phase, a received packet's state is calculated on the basis of almost the same workflow, and the algorithm confirms whether it is included in the state transition pattern of the normal communication model. If the transition is not included in the normal model, then a packet is regarded as anomalous.

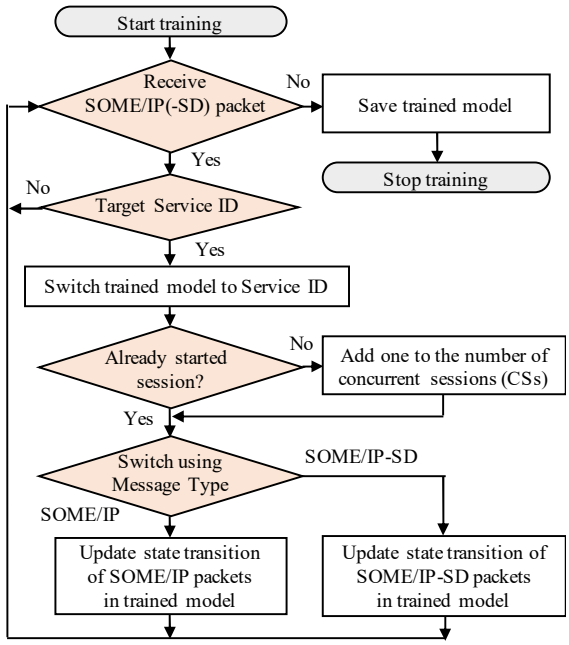

Fig. 7. Workflow for training phase of real-time detection algorithm.

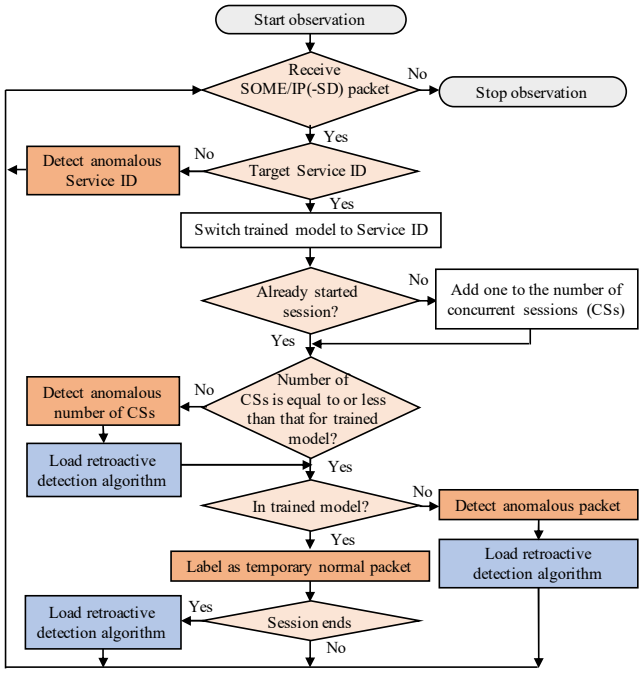

Fig. 8. Workflow for observation phase of real-time detection algorithm.

1) Training Phase: To generate the normal communication model, the following information is recorded as a state transition pattern. The first feature is the number of concurrent service sessions, that is, the number of concurrent service subscription ECUs. This corresponds to "Add one to the number of concurrent sessions (CSs)" in Fig. 7. As compared to communication in a general IT network, the number of ECUs receiving packets for a service here is relatively static. Thus,

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when an attacker transmits anomalous SOME/IP packets, the number of concurrent service sessions exceeds the learned value, and the attack can be detected. In terms of the SOME/IP specification, the Session-ID in the header (for SOME/IP-SD, as well) is counted one by one for each SOME/IP Message Type and each sender. By checking the Session-ID continuity, the number of simultaneous sessions can be obtained. If an attacker ignores the Session-ID order, i.e., inserts communication that is not based on the SOME/IP specification, the resulting Session-ID order error can easily be detected.

Other features recorded to form the state transition pattern are the communication interval between packets, the payload value, and the amount of change. These features are recorded for each type of packet. Table IV provides an example of a list of features recorded for a SOME/IP Response packet. A similar state transition is recorded for each of the other SOME/IP Message Types and SOME/IP-SD Entries. The communication interval is learned by discretizing it to some extent. For example, given the longest interval, shortest interval, and median value among the learned values, the communication interval is discretized into three values of short, medium, and long intervals. Next, because a designer can freely set the payload features for each service in SOME/IP, the normal model size may become excessively large if the payload values and changes are learned directly. Thus, the transition pattern of a payload is abstracted to reduce the amount of learning data. When the bit assignment specification is known from the design document, this information can be used. Otherwise, when the specification is unknown, an existing technique like that in [14] for estimating the bit assignment of CAN communication can also be applied to SOME/IP. Finally, when the payload includes a continuous value such as a vehicle speed, the difference between payloads with the continuous value (ΔP) is discretized into two values of "with/without change," because there are many types of payload variations.

2) Observation and Detection Phase: As shown in Fig. 8, each time a packet is received, features recorded in the training phase are calculated to confirm whether the packet is included in the normal communication model. If it is not included, then the packet is regarded as anomalous.

C. Retroactive Detection Algorithm

The retroactive detection algorithm detects anomalies that the real-time detection algorithm overlooked. As shown in Fig. 8, there are three situations in which the retroactive detection algorithm is executed.

1) Detection of an Anomalous Number of Concurrent Sessions (CSs): In-vehicle networks are static as compared to general IT systems, and the number of ECUs subscribing to a SOME/IP service is unlikely to increase or decrease frequently. Accordingly, if the number of CSs is higher than at the time when the normal model was learned, the IDS assumes that a "normal & anomaly" or MITM attack is occurring. At this timing, however, it cannot determine which packets are anomalous.

2) Real-Time Detection of an Anomalous Packet: When the real-time algorithm detects an anomalous packet, the results of all past packets received from the same sender are overwritten as anomalous. Then, all future packets received from the same sender are also regarded as anomalous.

TABLE IV.
LIST OF FEATURES RECORDED FOR SOME/IP RESPONSE PACKETS

Feature	Overview
$\Delta T_{prev.\ Req.}$	Communication interval between present Response and previous Request
$\Delta T_{prev.\ Res.}$	Communication interval between present Response and previous Response
A. $\Delta T_{prev.\ Req.}$	Average of $\Delta T_{prev.\ Req.}$
A. $\Delta T_{prev.\ Res}$	Average of $\Delta T_{prev.\ Req.}$
$\Delta P_{prev.\ Req.}$	Difference in payloads between present Response and previous Request
$\Delta P_{prev.\ Res.}$	Difference in payloads between present Response and previous Response
# of CSs	Number of concurrent SOME/IP sessions
OS	Order of Session-ID values

3) End of a Service Session: This workflow proceeds when an anomalous number of CSs is detected but the real-time detection algorithm did not detect any anomalous packets. In this case, certain sessions are anomalous but all of the packet transitions appear genuine. The retroactive detection algorithm thus computes the long-term time characteristics of each session. By comparing the average time interval of the packets in each session to that of the normal model, it regards sessions that deviate from the normal model and exceed the normal number of CSs as anomalous.

VI. EXPERIMENTAL RESULTS AND EVALUATION

For five of the nine attack patterns listed in Table III, the proposed IDS had a TPR of 1.0 and an FPR of 0.0. For the four remaining attack patterns, the IDS overlooked anomalous sessions and mistakenly detected normal sessions. Overall, the IDS had a TPR of approximately 0.91 and an FPR of 0.052. Of the packets regarded as anomalous, 18% were detected by the real-time detection algorithm. The remaining 82% were overlooked by the real-time algorithm but were detected by the retroactive algorithm. This was because most of the attack packets were highly similar to normal packets in terms of the features used by the real-time detection algorithm.

There were three types of attack patterns for which overlooked or erroneous detection occurred. The first error type occurred when a normal request-side ECU's communication was interrupted from start to finish because of an attack outside SOME/IP in a Request/Response service. This type of attack pattern is included in Anomaly_RR (b). When an attacker and a normal server participated in natural, indistinguishable communication, all of the attacker's anomalous packets were overlooked. The second error type occurred in Request/Response services when an attacker mediated a normal request as an MITM attack but stopped transferring genuine packets and sent plausible fake packets; in other words, the attacker performed impersonation. As a result, the IDS overlooked the fake packets. This type of attack pattern is in Anomaly_RR (a). The third error type occurred in Notification Events services. In this pattern, the attacker first impersonated a normal client ECU and a server to establish a session. Then, the attacker sent communications that were natural and indistinguishable by real-time detection from the start to the end of the session. As

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a result, the client ECU did not actually subscribe to the normal server's service, the attack packets were overlooked, and the normal server packets were falsely detected. This type of attack pattern is in Anomaly_NEpc (c) and Anomaly_NEc (c).

VII. Discussion

A. Normal and Attacker Models

We did not target the Fire & Forget and Fields services in this paper for two reasons. First, when a user tries to implement transmission patterns for control communication as in the CAN, it is feasible to use Request/Response and Notification Events communication. Second, as explained above, we can regard Fire & Forget and Fields as subsets of Request/Response and Notification Events, respectively. In other words, Fire & Forget can be treated as having a similar communication pattern simply by reversing the source and destination for Notification Events. In addition, Fields can be assumed as a combination of Request/Response and Notification Events. Fire & Forget and Fields can thus be used to obtain similar results by detecting anomalies with similar attack patterns. Thus, we focused on the Request/Response and Notification Events services as the normal communication patterns in this paper. However, it remains an open problem to confirm that there are no specific attack patterns for the Fire & Forget and Fields services. In addition, stronger attackers may exist. For example, multiple attacks can be performed simultaneously from multiple attack surfaces, such as the spoofing attack discussed here and an IP-based denial-of-service attack. Such stronger attacks are also open problems for the proposed IDS.

The datasets did not include use cases for Request/Response communication without establishing publish/subscribe. Our IDS might still able to observe such use cases, because most anomalous SOME/IP-SD packets were detected correctly by the retroactive detection algorithm from the result of a corresponding SOME/IP service. Furthermore, these use cases still transmit FindService and OfferService packets, which are enough to construct a meaningful normal model.

B. False Positives and False Negatives

The anomalous packets missed by the IDS represented communication in which no anomaly was found in the payload's timing or transition. In some attacks, the IDS can use lower-layer features such as an IP address or MAC address. However, if an attacker falsifies such features by manipulating network settings, the numbers of errors and misses may increase in other cases. Additional approaches would thus be useful to detect these natural, indistinguishable attacks. It may be possible to detect attacks in combination with technologies besides anomaly detection via packet information from layer 2 or higher. For example, we could combine the falsification detection of ECU firmware and a physical-layer-based source identification technique proposed for the CAN protocol.

VIII. Conclusion

In this paper, we classified normal communication patterns and attack patterns for SOME/IP to evaluate an IDS. We also proposed an anomaly detection system for detecting attacks. By combining two algorithms for real-time and retroactive detection, we confirmed that the proposed IDS could detect anomalous SOME/IP packets in a series of illegal packets transmitted by an attacker. Overall, the proposed IDS had a TPR of 0.91 and an FPR of 0.052. Here, we assumed that the SOME/IP packets were fixed-length and undivided. In the future, we will improve the proposed IDS by preparing normal communication patterns to account for cases without publish/subscribe, divided packets, and other service interfaces. In addition, for attack patterns that are difficult to detect by only monitoring SOME/IP packets, we will improve the system by integrating other features.

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